

INDEX + MATCH + MATCH in Excel

1. What is INDEX + MATCH + MATCH?

This is a two-way lookup. It finds the value where a row and a column cross in a table. One MATCH finds the row position, a second MATCH finds the column position, and INDEX returns the value at that crossing. It solves the problem of finding a number in a grid, such as the sales of one product in one region. A Data Analyst uses it to build flexible lookups that work even when rows or columns are added or moved.

2. Syntax

=INDEX(return_range, MATCH(row_value, row_range, 0), MATCH(column_value, column_range, 0))

Argument	Required?	What it accepts
return_range	Required	The table body only, without headings or row labels (for example B2:D5)
row_value	Required	The row label to find, such as "East" or a cell reference
row_range	Required	The column of row labels (for example A2:A5). It must have the same number of rows as return_range.
column_value	Required	The column heading to find, such as "Laptop" or a cell reference
column_range	Required	The row of headings (for example B1:D1). It must have the same number of columns as return_range.

The first MATCH gives the row number and the second gives the column number, always in that order. The 0 at the end of each MATCH means exact match. Always include it.

3. Simple Example

Find the Laptop sales for the East region:

	A	B	C	D	E	F	G	H
1	Region	Laptop	Mouse	Keyboard		Region	Product	Sales
2	North	1200	150	300		East	Laptop	1000
3	South	800	120	250				
4	East	1000	140	280				
5	West	900	110	220				

Formula in H2: =INDEX(B2:D5, MATCH(F2, A2:A5, 0), MATCH(G2, B1:D1, 0))

The first MATCH finds East in A2:A5: position 3. The second MATCH finds Laptop in B1:D1: position 1. INDEX returns the value at row 3, column 1 of B2:D5. **Result: 1000**

4. Another Example

Find the sales of one employee in one month:

	A	B	C	D	E	F	G	H
1	Name	Jan	Feb	Mar		Name	Month	Sales
2	Ayesha	5000	5200	5100		Tanvir	Feb	3900
3	Rahim	4200	4500	4400				
4	Nabila	6100	6000	6300				
5	Tanvir	3800	3900	4000				

Formula in H2: =INDEX(B2:D5, MATCH(F2, A2:A5, 0), MATCH(G2, B1:D1, 0))

The first MATCH finds Tanvir in A2:A5: position 4. The second MATCH finds Feb in B1:D1: position 2. INDEX returns the value at row 4, column 2 of B2:D5. **Result: 3900**

5. When to Use It

- Find sales by region and product, or by employee and month
- Look up a price from a price list using item and size
- Find stock levels by item and warehouse
- Find a score by student and subject
- Build interactive reports where the user changes the row and column values

6. Important Things to Remember

- The first MATCH finds the row and the second MATCH finds the column. The order inside INDEX is row, then column.
- return_range holds only the data. Do not include the headings or row labels.
- row_range and column_range must line up with return_range, with the same number of rows and columns.
- Use 0 in both MATCH functions for an exact match. If it is left out, MATCH uses 1 and can return wrong results.
- If a label is not found, the result is #N/A. Wrap the formula in IFERROR for a message.
- This formula works in every version of Excel, and labels are not case-sensitive.

7. Common Mistakes

Mistake: Swapping the two MATCH functions, which gives a wrong value or an error. **Fix:** Put the row MATCH first and the column MATCH second.

Mistake: Including the headings in return_range, which shifts every position by one. **Fix:** Start return_range at the first data cell.

Mistake: Ranges that do not line up, such as A2:A5 with B2:D6. **Fix:** Use the same rows for row_range and return_range.

Mistake: Leaving out the 0, so MATCH looks for an approximate match. **Fix:** Always end each MATCH with , 0).

8. Data Analyst Tip

When rows are identified by two labels, such as Region and Product with quarters across the top, match both with MATCH(1, (A2:A7="East")*(B2:B7="Laptop"), 0). The full formula is =INDEX(C2:F7, MATCH(1, (A2:A7="East")*(B2:B7="Laptop"), 0), MATCH("Q2", C1:F1, 0)). In Excel 2019 or earlier, confirm it with Ctrl + Shift + Enter.